Halving generalized cubes

Human Compression Project

Abstract

We apply the common graph-halving theorem to Danzer's generalized cubes. The automorphism group is the augmentation module of the vertex set extended by the original polytope group. This description recovers the polygon parameters and all locally spherical cases in the stated construction. Explicit Gray coordinates identify both the original and halved toroidal face posets, distinguished generators and translation lattices. These recover the principal conclusions of Montero and Ivić Weiss for their first arXiv edition, with its auxiliary rank qualification and exclusions stated separately.

1 The common halving theorem and source conventions

The common paper [U] proves the endpoint intersection criterion C.Endpoint and realizes the operation geometrically in C.Graph and C.Halving. In particular, for a regular lattice polytope, replacing an endpoint marker a by aba gives a regular hypertope. Its group is normal of index at most two; the index is two exactly when the relevant vertex graph is bipartite. If the old endpoint order is $m < \infty$, the new order is $m/\gcd(m, 2)$, and infinite order stays infinite. The other labels incident with b are duplicated. Reversing the types gives the source's convention at the final endpoint.

These are [MW20, Theorem 3.1, Proposition 3.2 and Corollary 3.3], with the same lattice hypothesis. The term *non-degenerate* is not replaced by an assumption about a diagram alone. The source's Lemma 2.4 follows directly because a subgroup generated by disjoint marked factors is their product and intersections are coordinatewise. Its quoted Proposition 2.1 is background; the incidence recognition needed from its quoted Theorem 2.2 is proved as UC.Geometry in [U]. The quoted Proposition 2.3 is not needed.

2 Parity modules and locally spherical geometries

Retain the following Danzer construction. If K is a finite regular lattice polytope with m vertices, 2^K has binary vertex set $V = F_2^m$. A face associated with a face F of K and $x \in V$ allows exactly the coordinates indexed by the vertices of F to vary. These coordinate subcubes, ordered by inclusion with their improper faces, form a regular lattice polytope of one higher rank, with vertex figures K and automorphism group

$$\mathrm{Aut}(2^K) = V \rtimes \mathrm{Aut}(K). \tag{M1}$$

The coordinate action is the natural vertex permutation action. This construction theorem is a substantial retained input; neither its polytopality nor the identification of the full group in (M1) is inferred solely from an order calculation.

Corollary C.Parity. Halving 2^K at its vertex endpoint gives the exact group

$$E_m \rtimes \mathrm{Aut}(K), \qquad E_m = \{x \in \mathbf{F}_2^m : \sum x_i = 0\}, \tag{M2}$$

of order $2^{m-1}|\mathrm{Aut}(K)|$, acting as the full type-preserving group of the regular hypertope in Corollary C.Halving.

Proof. The vertex graph of 2^K is the m-cube, bipartitioned by coordinate parity. Its color-preserving subgroup of (M1) is exactly (M2), so Corollary C.Halving identifies the whole halving group. Alternatively, let a flip the coordinate of a base vertex v of K and let b be its first distinguished reflection, carrying v to an adjacent vertex v'. The product $a'b = (aba)b$ is translation by $e_v + e_{v'}$. The unchanged reflections generate Aut(K). The original regular group is transitive on edges. The common graph-generation corollary in [U] therefore gives the whole augmentation module E_m, and proves generation of the displayed semidirect product. □

The source uses the dual convention $\widehat{P} = (2^{K^*})^*$. Consequently m in its formulas is the number of facets of K, and reversing all types converts (M2) exactly to its construction.

Corollary C.Polygon (polygon inputs). For a p-gon K, $p \geq 3$, the result is a self-dual orientable regular map of type $\{p, p\}$, with

$$V = F = 2^{p-1}, \qquad E = p2^{p-2}, \qquad |\operatorname{Aut}| = p2^p, \qquad g = 1 + 2^{p-3}(p-4). \tag{M3}$$

Proof. The two vertex colors give V and F after ordering the map types as (color 0, old square, color 1). Each edge of the p-gon specifies 2^{p-2} coordinate squares, giving E. The fork diagram becomes {p,p}. A single coordinate flip conjugates the two new endpoint reflections and exchanges the colors, supplying the duality. Extend the reflection-sign character of the dihedral Aut(K) trivially over E_m. Each of the three new distinguished reflections has sign -1, so the chamber orientation is well defined and reverses across every adjacency. This proves orientability. Euler's formula $V - E + F = 2 - 2g$ gives (M3). □

For $p = 3$ the map is the tetrahedron: the group is $E_3 \rtimes S_3$ of order 24 with the spherical A_3 marking and its universal Coxeter geometry. For $p = 4$ it is the square toroid conventionally denoted $\{4, 4\}_{2,2}$. The following toroidal section proves that lattice identification and identifies the original dual generalized cube with $\{4, 4\}_{4,0}$ by explicit face-poset and marking maps.

Corollary C.Spherical (all locally spherical inputs and exceptional orders). In output rank $n \geq 4$, the locally spherical cases of the source's dual generalized-cube construction are obtained from K equal to the $(n-1)$-simplex or $(n-1)-cube$, and additionally the dodecahedron for $n = 4$ and the 120-cell for $n = 5$. Their types are respectively spherical D_n, affine $\widetilde{B}_{n-1}$, and the rank-four/rank-five compact-hyperbolic forks. The last two finite groups have orders

$$2^{11} \cdot 120 = 245760, \qquad 2^{119} \cdot 14400. \tag{M4}$$

Proof. Put $L = K^*$ to use the coordinate model. Each of its two new vertex residues is L itself. If the output is locally spherical, L must therefore be a universal spherical polytope. Retain the classical finite string-Coxeter list. The lattice hypothesis excludes digonal rank-two sections, so every relevant adjacent label is at least three. In rank at least four a proper three-node fork with equal labels q,q is spherical only for $q = 3$, where q is the first label of L. In rank five, the remaining candidate $F_4 = \{3, 4, 3\}$ is excluded because its proper four-node fork is affine. The finite list leaves exactly the simplex, cross-polytope, icosahedron and 600-cell, the last two only in their respective ranks. For rank at least six only the first two remain.

Conversely, all proper faces of these L are simplices. At an old j-face of 2^L, $j \geq 2$, let F be its associated $(j-1)$-face of L. The lower and upper sections vary independently. After halving, the maximal residue is the product of the compressed cube 2^F with the spherical section L/F. As F has j vertices, the first factor has group $E_j \rtimes S_j$ and its D_j marking; its order $2^{j-1}j!$ equals that of the finite Coxeter group D_j, proving equality with the universal spherical geometry. Here $D_2 = A_1 \times A_1$ and $D_3 = A_3$. Vertex residues are L, so all maximal residues, and consequently all proper residues, are genuinely spherical geometries. A diagram alone is not used as a substitute for those residues.

The new global diagrams follow from Corollary C.Halving. The retained finite, affine and compact-hyperbolic Coxeter classifications identify them as stated. Formula (M2), with $m = 12$ or 120 and the classical orders $|\operatorname{Aut}(dodecahedron)| = 120$ and $|\operatorname{Aut}(120-cell)| = 14400$, gives (M4). For the simplex $m = n$ and $|\operatorname{Aut}(K)| = n!$, giving D_n order $2^{n-1}n!$. For the $(n-1)-cube$, $m = 2(n-1)$ and

$|\operatorname{Aut}(K)| = 2^{n-1}(n-1)!$, giving $2^{3n-4}(n-1)!$. The finite hyperbolic examples are proper hypertopes because the fork diagrams have a vertex of degree three. □

The exact all-rank toroidal labels require more than these orders and residues. They are established in the next section by an explicit coordinate identification.

3 Exact toroidal face posets and translation lattices

Let $r \geq 2$ and let L be the r-dimensional cross-polytope with vertices labelled $+e_i, -e_i$, $1 \leq i \leq r$. A proper face of L chooses at most one vertex from each antipodal pair. Write Q_r for the cubical tessellation of $\mathbb{R}^r$ whose vertices are $\mathbb{Z}^r$ and whose cells are products of unit intervals and integer singletons. The quotient $Q_r/4\mathbb{Z}^r$ has a formal least and greatest face adjoined when viewed as an abstract polytope. These formal faces are also understood below.

For each pair of binary coordinates belonging to $+e_i, -e_i$, use the Gray map

$$(0,0) \mapsto 0, \quad (1,0) \mapsto 1, \quad (1,1) \mapsto 2, \quad (0,1) \mapsto 3 \quad \text{in } \mathbb{Z}/4\mathbb{Z}. \tag{T1}$$

Lemma C.Gray. The product of these maps is a rank-preserving face-poset isomorphism $2^L \cong Q_r/4\mathbb{Z}^r$. Binary weight modulo two becomes $\sum_i x_i$ modulo two in the cubical coordinates.

Proof. In Danzer's construction a proper face of 2^L allows the bits indexed by a face of L to vary, and fixes every other bit. Thus in any antipodal pair either both bits are fixed or exactly one bit varies. Under (T1), varying a single bit describes precisely one edge of the four-cycle: its four possible edges arise from the two choices of varying bit and the two values of the fixed bit. A face therefore becomes a product of four-cycle vertices and edges. Conversely every such product uniquely specifies the varying bit in each free pair and the remaining fixed bits. Its dimension is the number of free pairs. Inclusion is preserved and reflected coordinatewise, and formal faces match. These products are exactly the cubical cells modulo $4\mathbb{Z}^r$, with their incidence. The four cases of (T1) have binary parities 0,1,0,1 respectively, proving the last assertion. □

This is a face-poset identification, rather than an inference from orders. It also handles the dual in the target source explicitly. For a cubical face F with free coordinate set S, write its free intervals as $[m_i, m_i + 1]$ for $i \in S$ and its fixed coordinates as m_j for $j \notin S$. The dual face F^* has fixed coordinate $m_i + 1/2$ for $i \in S$ and free interval $[m_j - 1/2, m_j + 1/2]$ for $j \notin S$. This reverses containment: whenever an endpoint is fixed in a subface of F, the corresponding free interval of the dual contains the fixed midpoint of F^*. It is a bijection onto the shifted cubical tessellation, with dimensions adding to r. Subtracting $(1/2, \ldots, 1/2)$ identifies that tessellation with Q_r. The construction commutes with translations by $4\mathbb{Z}^r$, and consequently induces an order-reversing bijection on the quotient. Thus for the r-cube $K = L^*$, the source's polytope $(2^{K^*})^*$ is indeed the cubical toroid $\{4, 3^{r-2}, 4\}_{(4,0,\ldots,0)}$. The string is $\{4,4\}$ when $r = 2$. Reversing the source marking moves its last-end halving to first-end halving of 2^L. Type reversal is kept explicit; no unproved compatibility of a chosen self-duality with halving is needed.

3.1 Markings and the affine quotient

Use the following reflections of $\mathbb{R}^r$:

$$\begin{aligned} a(x_1, x_2, \ldots, x_r) &= (1 - x_1, x_2, \ldots, x_r), \\ b_i \ : \ x_i &\leftrightarrow x_{i+1} \quad (1 \leq i < r), \\ c(x_1, \ldots, x_{r-1}, x_r) &= (x_1, \ldots, x_{r-1}, -x_r). \end{aligned} \tag{T2}$$

They generate $W = \mathbb{Z}^r \rtimes B_r$, where B_r is the signed permutation group. Indeed the b_i, c generate B_r, including the reflection $x_1 \mapsto -x_1$; its product with a is translation by $\pm e_1$. Signed permutations then supply every integer translation. The simplex $1/2 \geq x_1 \geq x_2 \geq \cdots \geq x_r \geq 0$ is a fundamental

chamber. One way to verify this without a presentation theorem is to translate each coordinate to $[-1/2, 1/2]$, make it nonnegative, and order the coordinates; a point with strict inequalities has a unique such representative. Its reflected walls are the displayed generators. Its vertices are the barycentres of a nested flag of cubical faces, so its chamber system is the barycentric subdivision of Q_r.

The marking in (T2) also agrees with Danzer's marking under (T1). Flipping the $+e_1$ bit induces $x_1 \mapsto 1 - x_1$ modulo four; interchanging antipodal pairs induces the b_i; swapping the two bits in the last pair induces $x_r \mapsto -x_r$. These are the usual base vertex/base flag generators of 2^L. Hence the isomorphism of Lemma C.Gray identifies the marked action, not only the unmarked face poset. The action of W on $Q_r/4\mathbb{Z}^r$ has kernel exactly $\Lambda = 4\mathbb{Z}^r$: an affine signed permutation acting trivially on all vertices has translation zero modulo four, and its linear part is the identity modulo four; testing the unit vectors forces that signed permutation to be the identity.

Put $b = b_1$ and replace a by

$$d = aba : \ (x_1, x_2, x_3, \dots, x_r) \mapsto (1 - x_2, 1 - x_1, x_3, \dots, x_r). \tag{T3}$$

Let $E = \{v \in \mathbb{Z}^r : \sum v_i \text{ is even}\}$.

Lemma C.Affine. $H = \langle d, b_1, \dots, b_{r-1}, c\rangle$ is exactly $E \rtimes B_r$. It has index two in W and contains Λ. Its fundamental chamber is

$$x_1 \geq x_2 \geq \cdots \geq x_r \geq 0, \qquad x_1 + x_2 \leq 1. \tag{T4}$$

Its chamber system is the halved cubical geometry, and the marked finite halving group and geometry are H/Λ and the quotient of that chamber system by Λ, respectively.

Proof. Every displayed generator preserves the parity of an integral vertex's coordinate sum, so $H \leq E \rtimes B_r$. It contains B_r. The product of d with the reflection in B_r $(x_1, x_2) \mapsto (-x_2, -x_1)$ is translation by $e_1 + e_2$. Conjugating by B_r gives every $\pm e_i \pm e_j$. These span E: subtract multiples of $e_i - e_r$, $i < r$, to leave an even multiple of e_r, and use $2e_r = (e_r + e_1) + (e_r - e_1)$. This works also for $r = 2$. Thus equality holds and the index is two. Since every $4e_i$ is in E, $\Lambda \leq H$.

The region (T4) is the union of the preceding W chamber and its reflection in the a-wall $x_1 = 1/2$. On their common wall the two chambers glue; the other wall $x_1 = x_2$ is paired with its a-image $x_1 + x_2 = 1$. All other walls agree. Because $W = H \sqcup Ha$ and the interiors of W chambers are disjoint, these doubled chambers tile $\mathbb{R}^r$ under H, each once. Their wall reflections are exactly $d, b_1, \dots, b_{r-1}, c$. Equivalently, their chamber labels are the old cubical flags with the initial vertex chosen in one parity class, pairing the two old flags differing only at that vertex. The two ends of the old edge become the two new vertex types; the higher faces retain their type. This is precisely the graph-halving construction in [U].

Finally, translations in Λ preserve both vertex colors and take cubical faces to cubical faces. The same flag pairing therefore commutes with Λ. Every finite chamber has a lift, and two lifts yield the same finite chamber exactly when they differ by Λ. Wall adjacencies are the marked reflections modulo Λ. The maximal parabolics are the images of the corresponding wall subgroups, so the typed coset geometry also agrees. Regularity, thinness and residual connectedness of the finite geometry follow from the already proved halving theorem; they are not inferred solely from being a group quotient. □

For $r \geq 3$, the walls (T4) have the affine fork diagram $\widetilde{B}_r$: d and b_1 commute; both meet b_2 with product order three, successive b_i have order three, and $b_{r-1}c$ has order four. All other distinct wall pairs commute. This follows directly from (T2)–(T3), or from their mirror angles. Thus its rank-(r+1) quotient has the source's type $\{4, 3^{r-3}, 3/3\}$. Here the quotient lattice is exactly $4\mathbb{Z}^r$ in the original coordinates; the subscript $(4, 0, \dots, 0)$ denotes the lattice generated by that vector's signed-permutation orbit. This proves the source's all-rank toroidal identification, including the marking. Its group order is $[E : 4\mathbb{Z}^r]|B_r| = (4^r/2)2^r\, r! = 2^{3r-1} r!$, agreeing with the separate parity-module calculation. The order is a check, not the proof.

The coordinate convention can be checked directly against Ens's primary paper: its equations (3.1), (4.1) are (T2), (T3) for $r = 3$; its §2 definition of Λ_s uses the signed-permutation orbit of s, and

Theorem 4.3 denotes the corresponding Coxeter-complex quotient by that lattice. Thus $s = (4, 0, 0)$ here is literally the same parameter. The target extends this convention to larger r. No classification theorem about all invariant lattices is needed. See [Ens, pp.70–75]. Only that lattice convention is borrowed here, not its classification theorem.

3.2 The rank-three square map

When $r = 2$, reorder the halved types as (even vertex, old square, odd vertex). An old unit square joins its two even corners as an edge of the new map; the four even neighbors of an old odd vertex bound a new square face. This describes the ordinary square tessellation with vertex lattice E and edge directions

$$f_1 = (1, 1), \qquad f_2 = (-1, 1). \tag{T5}$$

Every even vertex has four such incident edges, every displayed quadrilateral is a face, and the incidences are exactly those of halving. Thus this is an isomorphism of maps, not a comparison of their counts. In this basis

$$4e_1 = 2f_1 - 2f_2, \qquad 4e_2 = 2f_1 + 2f_2.$$

The quotient lattice has basis $(2, -2), (2, 2)$ in square coordinates. Negating the first vector gives the customary pair $(-2, 2), (2, 2)$, so the map is $\{4, 4\}_{2,2}$. Lemma C.Gray and the explicit duality already identify the original map as $\{4, 4\}_{4,0}$. This proves both $p = 4$ identifications in [MW20].

4 Scope, rank qualification and attribution

The specified edition is arXiv:2001.11082v1, dated 29 January 2020, eleven physical pages. Its generalized-cube construction, polygon applications, complete locally spherical list and exact toroidal identifications are the conclusions recovered above. The printed names of toroidal geometries are justified by face-poset and marked lattice maps, not just group orders.

There is a rank qualification in Remark 4.1. Retain the Pellicer extension construction used in that remark. After endpoint-2s halving its three-node fork triangle has labels (p, p, s). In output rank at least four, with finite $p, s \geq 3$, this is a proper rank-three diagram and is not finite, since $2/p + 1/s \leq 1$. The result is therefore not locally spherical. In rank three all proper residues have rank at most two, and these finite parameters give finite dihedral residues. The unqualified wording consequently does not apply in rank three under the source's definition. No classification of the entire extension construction is asserted here.

The comparison against every example in the source's reference [9], and its unused alternating-semiregular identification, are excluded auxiliary assertions. Danzer's construction and full original automorphism group, classical Coxeter classifications and convex polytope orders, and standard orientation and Euler-characteristic facts are retained inputs. The selected pages of Ens fix the toroidal lattice convention; its classification theorem is not used to prove our particular quotient identifications.

The new use of the common binary module theorem is exact: conjugates of one edge difference generate its augmentation module, so the same generation argument serves this application and the equal-parameter polyhedra. It does not supply an independent proof of Danzer's polytopality theorem. The present application remains written rather than fully Lean-formalized; the common endpoint file verifies only its specified fixed-overlap identity.

References

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[MS02] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002.